LAB 02

Agile Backlog Creation & Sprint Simulation in Jira

Project: **Pharmacy Expiry & Re-order Dispatch Engine**

Objective	Convert pharmacy system functions into Epics and User Stories, prioritize and estimate the backlog, and plan two sprint simulations.
Actors	Pharmacy Clerk; Inventory Supplier; Pharmacy Administrator.
Source basis	Lab 03 UML use cases in the repository and the Lab 02 handout requirements.

Important evidence note

This document contains a proposed backlog, estimates, sprint allocation and reflection draft. It does **not** claim that Jira actions have already been performed. Capture the actual Jira backlog, Active Sprint board and Burndown Chart from your workspace and add those screenshots to the repository.

Project scope reflected in the UML

The Lab 03 UML identifies the Pharmacy Clerk and Pharmacy Administrator as system users, and the Inventory Supplier as an external participant. Its use cases include viewing medicine stock, dispensing medicine, generating an FEFO dispensing list, sending an automatic re-order purchase order, and managing safety-stock thresholds.

1. Epics and User Stories

EPIC 1 - Medicine Stock Visibility

Allow pharmacy staff to view current medicine inventory so they can make informed dispensing and replenishment decisions.

Story / Summary	User story	Priority	Points
LAB2-01 - View Medicine Stock	As a Pharmacy Clerk, I want to view the available stock of each medicine, so that I can check availability before dispensing.	High	3
LAB2-02 - View Medicine Details	As a Pharmacy Clerk, I want to view medicine details including batch and expiry information, so that I can identify the correct stock to dispense.	High	5

EPIC 2 - Safe Medicine Dispensing (FEFO)

Support safe dispensing by presenting stock in First-Expiry-First-Out (FEFO) order.

Story / Summary	User story	Priority	Points
LAB2-03 - Generate FEFO Dispensing List	As a Pharmacy Clerk, I want the system to generate a dispensing list ordered by earliest expiry date, so that medicines closest to expiry are dispensed first.	High	8
LAB2-04 - Dispense Medicine	As a Pharmacy Clerk, I want to record medicine dispensing against the selected batch, so that stock levels remain accurate after each transaction.	High	5

EPIC 3 - Automatic Re-order and Supplier Dispatch

Create and send purchase orders when inventory reaches the configured replenishment threshold.

Story / Summary	User story	Priority	Points
LAB2-05 - Detect Re-order Requirement	As a Pharmacy Administrator, I want the system to identify medicines that reach the re-order threshold, so that low stock can be replenished on time.	High	5
LAB2-06 - Generate and Send Purchase Order	As a Pharmacy Administrator, I want the system to generate and send a purchase order to the Inventory Supplier, so that replenishment can begin without manual order preparation.	High	8

EPIC 4 - Safety-stock Configuration

Let the Pharmacy Administrator maintain safety-stock thresholds used by the re-order process.

Story / Summary	User story	Priority	Points
LAB2-07 - Configure Safety-stock Threshold	As a Pharmacy Administrator, I want to set or update the safety-stock threshold for a medicine, so that re-ordering reflects the pharmacy's required minimum stock.	Medium	3
LAB2-08 - Review Threshold-based Stock Status	As a Pharmacy Administrator, I want to identify medicines at or below their safety-stock threshold, so that I can monitor items requiring attention.	Medium	3

2. Priorities and Estimation

The estimates use Fibonacci-style values listed by the Lab 02 handout. Three points represent straightforward view/configuration work; five points represent additional batch or inventory-state handling; eight points represent more involved decision logic or multi-step processing. These are proposed relative estimates and should be confirmed through a Planning Poker discussion.

Backlog total: 40 story points. **Sprint 1:** 21 points. **Sprint 2:** 19 points.

3. Two-sprint simulation plan

Sprint 1 - Stock visibility and safe dispensing
Goal: Enable the clerk to inspect available medicine and follow FEFO dispensing.

Story	Points	Planned completion
LAB2-01 - View Medicine Stock	3	Done (simulation target)
LAB2-02 - View Medicine Details	5	Done (simulation target)
LAB2-03 - Generate FEFO Dispensing List	8	Done (simulation target)
LAB2-04 - Dispense Medicine	5	Done (simulation target)

Sprint 2 - Re-order and safety-stock management
Goal: Support threshold-based replenishment and supplier purchase-order dispatch.

Story	Points	Planned completion
LAB2-05 - Detect Re-order Requirement	5	Done (simulation target)
LAB2-06 - Generate and Send Purchase Order	8	Done (simulation target)
LAB2-07 - Configure Safety-stock Threshold	3	Done (simulation target)
LAB2-08 - Review Threshold-based Stock Status	3	Done (simulation target)

The handout specifies one-week sprints. In Jira, create each sprint, set its goal, start it, move each story through To Do, In Progress and Done, and complete the sprint. These tables are a plan for the simulation, not evidence that the work has been completed.

4. Reflection questions - draft answers

1. Did your estimations reflect the actual effort?
The estimates provided a useful comparison of relative effort. Stock-view and threshold-configuration stories were estimated at 3 points, while FEFO generation and purchase-order dispatch received higher estimates because they involve more decision logic and validation. Since this was a short simulation rather than a completed implementation, the estimates indicate expected complexity and may differ from actual development effort.

2. Was your backlog well-prioritized?
The backlog placed stock visibility and safe dispensing first because the clerk needs reliable stock and expiry information before dispensing. Re-order detection and purchase-order dispatch followed, while threshold configuration and review were included to support replenishment. This ordering reflects the main user workflows and dependencies between them.

4. Reflection questions - continued

3. How did your simulated sprint align with your plan?
The plan divided the work into two focused sprints. Sprint 1 covered stock visibility and FEFO-based dispensing. Sprint 2 covered automatic re-ordering, supplier dispatch and safety-stock management. Update this answer after running the Jira simulation to state whether each story reached Done within its sprint and whether any items remained incomplete.

4. What insights did the burndown chart give about your team's capacity?
The burndown chart compares remaining estimated work with an ideal progress line. A steady decline suggests work is progressing consistently; a flat section suggests little work was completed during that period; and a sudden drop can indicate several stories were completed together or that work was updated late. The chart helps assess whether the sprint commitment was realistic, but a short classroom simulation is not enough to establish long-term team velocity.

5. Jira evidence to attach

Evidence	Screenshot file name
Backlog with Epics and Stories	screenshots/01_backlog_epics_stories.png
Story point assignments and priorities	screenshots/02_story_points_priorities.png
Sprint 1 Active Sprint board	screenshots/03_sprint1_active_board.png
Sprint 2 Active Sprint board	screenshots/04_sprint2_active_board.png
Completed sprint Burndown Chart	screenshots/05_burndown_chart.png

The burndown chart must be captured from Jira after story points and status transitions have been recorded. Do not substitute a manually drawn or invented chart.

Source and scope note

The Lab 03 UML file was readable and provided the named actors and use cases used here. The Lab 01 requirements spreadsheet and PDF are present in the repository, but their binary contents could not be decoded by the available GitHub file reader during preparation. Review this backlog against those Lab 01 details and adjust it if they specify additional functional requirements.